# TaSain Valdez Thomas

Software Engineer (AI/ML)
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## SUMMARY

Software engineer with hands-on ML depth. Builds production software in Python, C++, and SQL -- a worker-matching engine adopted by a manufacturing plant, a flutter PDF editor for a real client, and a phishing-campaign simulator with JWT auth and MySQL analytics. Trained a SE-DenseNet121 (PyTorch) with split conformal prediction: 90% prediction intervals achieving 90.4% empirical coverage and 2.81 BMI-point MAE (11.56% MAPE) on a held-out test set. Seeking a software or ML engineering role where rigorous engineering and honest validation matter.

## EDUCATION

**Wayne State University, James and Patricia Anderson College of Engineering** **May 2026**
*Bachelor of Science, Computer Science* (GPA: 3.4/4.0) *Detroit, MI*

- **Coursework:** LoRA, QLoRA, RLHF, DPO, GRPO, Red Teaming, Machine Unlearning, Grounded VLMs, Agentic AI Safety, Chain-of-Thought Reasoning, HarmBench, SafeChain

## CORE COMPETENCIES

**CORE COMPETENCIES**: Software engineering, Machine learning, Data pipelines, API development, Algorithms & data structures, Uncertainty quantification & conformal prediction, Model Validation, AI agent orchestration, Technical leadership
**TECHNICAL SKILLS**: Python, C++, SQL, SQLite, JavaScript, Dart, PyTorch, Conformal Prediction, LLM-as-Judge, GPT-4.1-Nano, Docker, Proxmox, GPU Passthrough, DigitalOcean, Nginx, systemd, Stripe, Telegram Bot API, Git/GitHub Workflow (GitFlow), Inductive Automation Ignition, Vuforia Model Targets

## PROFESSIONAL EXPERIENCE

**Industry 4.0 / Automation Intern (Rotational)** | ***Magna EV Structures*** **Sep 2023 - Sep 2024**

- Built a worker-to-station assignment engine (C++/SQLite) using a weighted evaluateFit() scoring function that balanced line familiarity, station expertise, and rank - adopted by the plant to auto-assign workers during real shift changes.
- Developed an AR-based vision inspection system for a tablet that performed model-based screw-position recognition and pass/fail logging using Vuforia Model Targets, Inductive Automation Ignition, CSV data pipelines, and Python API scripts.
- Built managerial monitoring systems tracking production output, attendance, and custom filtering using role-based authorization (supervisor vs manager tiers), with least-privilege write access and audit logging of manager changes to the test database.

## ENGINEERING PROJECTS

**ConformalBMI: Uncertainty-Aware Body Mass Estimation from Images (Python, PyTorch)** **Apr 2026 - Present**
GitHub

- Translated published BMI-estimation research (Digital Scale; Sui et al.; Manichand et al.; Jin et al.) into a working SE-DenseNet121 (PyTorch) estimator, trained on Celeb-FBI (7,208 real images, 6,174 labeled after curation). Reaching 2.81 BMI-point MAE (11.56% MAPE) on a 620-sample held-out test set.
- Extended the model with split conformal prediction for distribution-free uncertainty: 90% prediction intervals (q = 6.44) with 90.4% empirical coverage on 620 held-out samples.
- Layered a GPT 4.1-Nano LLM-as-judge reasoning module that explains each estimate in plain English and flags high-uncertainty and extreme-BMI tail cases.

**Phishing Campaign Simulator (Security Awareness) - Node.js, MySQL, HTML/CSS/JS** **Jan 2025 - May 2025**
GitHub
***Wayne State University***

- Built a phishing-campaign simulator that sends templated emails over SMTP with a tracking pixel, logging open/click metrics in MySQL.
- Implemented JWT (jsonwebtoken) authentication with token-scoped separation of user vs admin views, plus least-privilege data access.
- Designed an analytics pipeline on a command design pattern, routing events to a formatting service.

**Client Project - PDF Editing & File Management (Flutter)** **Jan 2026 - May 2026**
***Wayne State University (NDA: tech + role only)***

- Led a 4-person team building a Flutter collaboration app (PDF annotation, video conferencing, subscriptions) for a real client. Ran 3 weekly meetings, client comms, and a GitFlow branching/PR workflow.
- Personally built the PDF editor; floating toolbar (pen/pencil/highlighter/eraser) layering annotations over a PDF and flattening to a single saved page, with an app-bar of undo/redo/delete/save and a text-insert command.
- Revised the in-app activity meter tracking user progress and time spent.

**Homelab - Privacy-First AI Inference** **Mar 2026 - Present**

- Self-hosted Proxmox hypervisor with GPU passthrough running an AI agent in a dedicated VM; assembled hardware and configured resource allocation.

**On-Wheels Detailing - Founder & AI Agent Orchestration** GitHub **May 2020 - Present**

- Founded and operate a detailing business with actual paying customers; wired up Stripe payment processing end to end.
- Built a Telegram Bot API integration for daily briefings and real-time lead alerts; orchestrated a self-hosted AI agent to generate the site under a no-customer-data-persisted constraint.